

Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study ¹
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1 Objective

To implement a Decision Tree classifier, extend it into a Random Forest ensemble model, study the impact of hyperparameters on overfitting and generalization, select optimal hyperparameters using 5-fold cross-validation, and compare the single-tree and ensemble-tree models on the Wisconsin Diagnostic Breast Cancer dataset.

2 Problem Statement

Given the Wisconsin Diagnostic Breast Cancer dataset of 30 numerical features computed from digitized images of breast mass fine-needle aspirates, the problem is binary classification: predict whether a tumour is malignant or benign. The objective is to determine how tree depth and other hyperparameters affect overfitting in a single Decision Tree, whether Random Forest's bagging and random feature selection improve generalization over a single tree, and which model is ultimately preferable for this diagnostic task.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
Dataset Source	UCI Machine Learning Repository – https://archive.ics.uci.edu (loaded via <code>sklearn.datasets.load_breast_cancer</code>)
Number of Samples	569
Number of Features	30 numerical attributes (mean, standard error, and “worst” value of 10 cell-nucleus measurements)
Number of Classes	2 (357 benign / 212 malignant)
Missing Values	0
Train-Test Split	80:20 stratified (455 train / 114 test), <code>random_state=42</code>

4 Exploratory Data Analysis

Include all relevant visualizations.

- Dataset overview

- Statistical summary
- Missing value analysis
- Class distribution
- Correlation matrix
- Feature distribution

Class Distribution

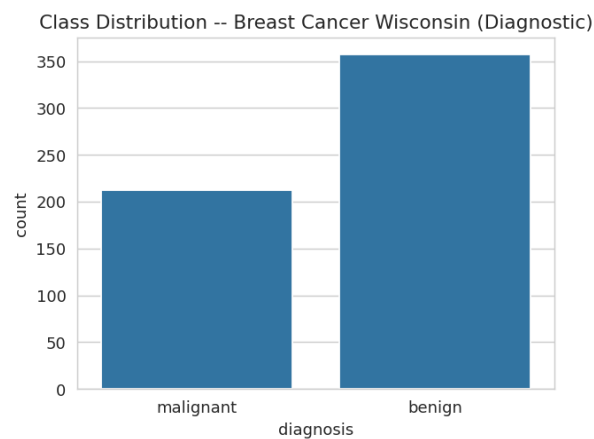


Figure 1: Class distribution – Breast Cancer Wisconsin (Diagnostic)

Inference: The dataset is mildly imbalanced (357 benign vs. 212 malignant, about 63:37). No missing values are present, so no missing-value visualization is required.

Correlation Matrix

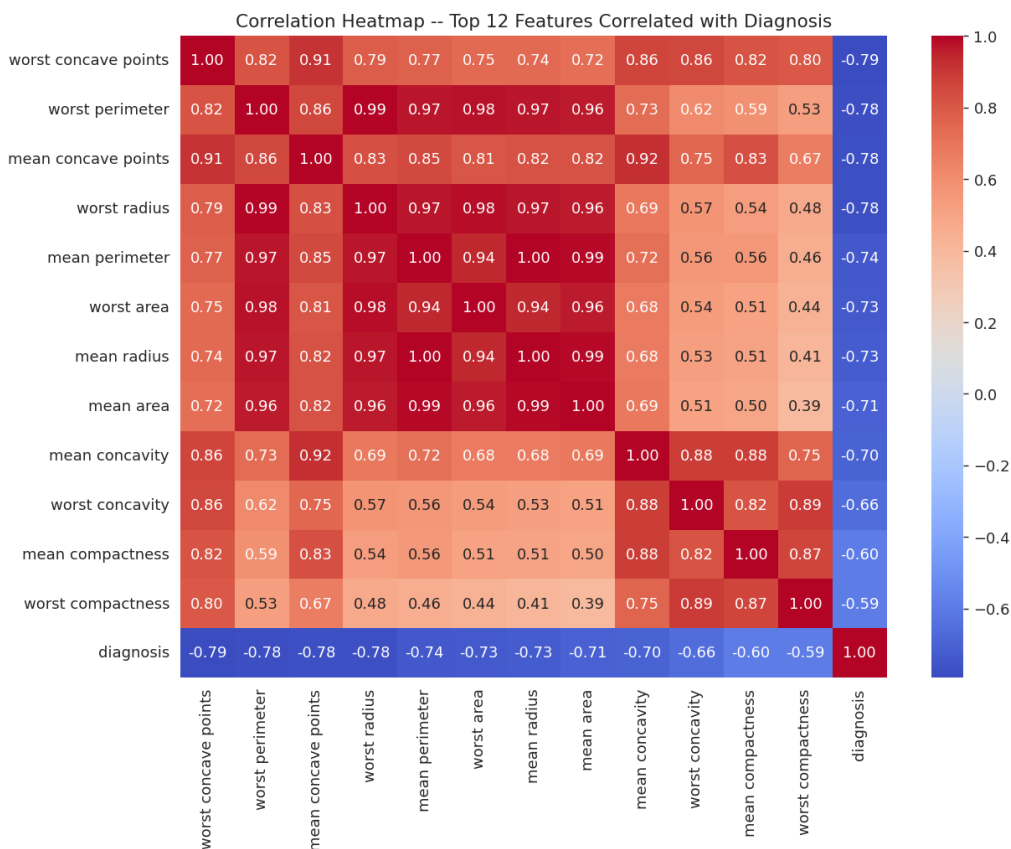


Figure 2: Correlation heatmap – top 12 features correlated with diagnosis

Inference: “Worst concave points”, “worst perimeter”, “worst radius”, and “mean concave points” show the strongest correlation with diagnosis (all $|r| > 0.7$), consistent with the fact that larger, more irregularly-shaped nuclei are a hallmark of malignancy. These same features later emerge as the top Random Forest feature importances (Section 12) and the root splits of the visualized decision tree (Section 13), corroborating each other.

5 Data Preprocessing

Explain:

- **Label encoding:** The target is already numerically encoded by scikit-learn (0 = malignant, 1 = benign); no additional encoding step was required.
- **Missing value handling:** Not required – the dataset contains zero missing values across all 569 rows and 30 features.
- **Feature scaling:** Not required for either model – Decision Trees and Random Forests split on individual feature thresholds and are invariant to monotonic feature scaling, unlike distance-based or gradient-based models.
- **Train-test split:** 80:20 stratified split (455 train / 114 test rows), preserving the 63:37 class ratio in both subsets.

6 Machine Learning Algorithm

Explain:

- **Working principle:** A Decision Tree recursively partitions the feature space by selecting, at each node, the feature and threshold that most reduces class impurity (Gini index or entropy), forming a hierarchy of if-then decision rules ending in leaf-node class predictions. Random Forest is an ensemble of many such trees, each trained on a bootstrap sample of the training data (bagging) and each considering only a random subset of features at every split; final predictions are made by majority vote across all trees.
- **Mathematical formulation:**

$$\text{Gini}(t) = 1 - \sum_c p(c | t)^2, \quad \text{Entropy}(t) = - \sum_c p(c | t) \log_2 p(c | t)$$

$$\hat{y} = \underset{i=1, \dots, B}{\text{mode}} T_i(x)$$

where T_i is the i -th tree trained on a bootstrap sample, and B is the number of trees (`n_estimators`).

- **Advantages:** A single Decision Tree is highly interpretable (Figure 13) and requires no feature scaling. Random Forest substantially reduces variance relative to a single tree by averaging over many decorrelated trees (via bagging and random feature selection), typically improving generalization.
- **Limitations:** A single Decision Tree is prone to high variance and overfitting, especially as depth grows unconstrained (Section 9). Random Forest loses the single tree's direct interpretability (though feature importances partially compensate, Section 12), requires more memory and training time to fit many trees, and its many hyperparameters increase tuning cost (Section 8).

7 Experimental Setup

Python Version	
Scikit-Learn Version	
IDE	Jupyter Notebook
Operating System	
Hardware	

8 Hyperparameter Selection

Model	Search Method	Best Parameters	Best CV Accuracy
Decision Tree	Grid Search	criterion=gini, max_depth=5, min_samples_split=2, min_samples_leaf=4	0.9385
Random Forest	Grid Search	n_estimators=100, max_depth=10, max_features=log2, bootstrap=True	0.9648

Note: Decision Tree was searched over $\text{criterion} \in \{gini, entropy\} \times \text{max_depth} \in \{2, 3, 4, 5, 7, 10, None\} \times \text{min_samples_split} \in \{2, 5, 10\} \times \text{min_samples_leaf} \in \{1, 2, 4\}$ (126 combinations). Random Forest was searched over $\text{n_estimators} \in \{50, 100, 200\} \times \text{max_depth} \in \{3, 5, 10, None\} \times \text{max_features} \in \{sqrt, log2\} \times \text{bootstrap} \in \{True, False\}$ (48 combinations). Both used 5-fold cross-validation optimizing accuracy.

Decision Tree Cross-Fold Results (summarized by criterion and max_depth)

Criterion	Max Depth	Avg CV Accuracy
gini	2	0.9311
gini	3	0.9333
gini	5	0.9385
gini	10	0.9233
gini	None	0.9226
entropy	2	0.9209
entropy	3	0.9333
entropy	5	0.9263
entropy	10	0.9275
entropy	None	0.9250

Inference: Accuracy is highest around `max_depth=3--5` for both criteria and noticeably lower once the tree is left unconstrained (`max_depth=None`), directly reflecting overfitting at high depth – consistent with the depth-vs-accuracy curve in Section 9. Gini and entropy perform similarly overall, with Gini marginally ahead at the selected depth.

Random Forest Cross-Fold Results (summarized by n_estimators, max_depth, max_features)

n_estimators	Max Depth	Max Features	Avg CV Accuracy
50	10	log2	0.9615
50	10	sqrt	0.9582
100	10	log2	0.9626

n_estimators	Max Depth	Max features	Avg CV Accuracy
100	10	sqrt	0.9604
100	None	log2	0.9648
200	10	log2	0.9615
200	None	log2	0.9626

Inference: Accuracy is fairly stable across n_estimators once past 50 trees, and max_features=log2 consistently edges out sqrt on this 30-feature dataset. Unlike the Decision Tree, deeper/unconstrained Random Forest trees do not hurt generalization – bagging’s variance reduction compensates for individual trees overfitting, which is the central theoretical advantage of the ensemble.

9 Results

Metric	Decision Tree (tuned)	Random Forest (tuned)
Accuracy	0.9035	0.9561
Precision	0.9420	0.9589
Recall	0.9028	0.9722
F1-score	0.9220	0.9655
ROC-AUC	0.9358	0.9924
Training Time (s)	0.0061	0.1489

10 Result Visualization

Include all graphs generated during the experiment.

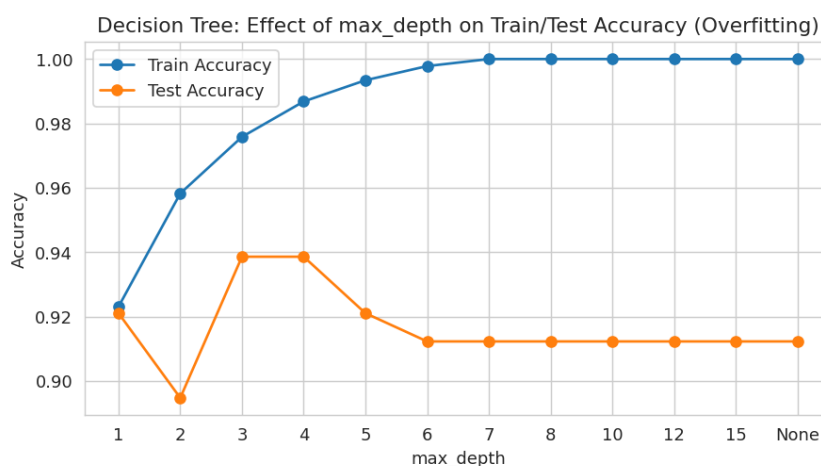


Figure 3: Decision Tree: effect of max_depth on train/test accuracy

Inference: Training accuracy climbs to a perfect 1.0 by max_depth=7 and stays there, while test accuracy peaks around max_depth=3--4 (≈ 0.939) and then *declines* to

a lower plateau (≈ 0.912) as depth increases further – a textbook overfitting curve: the tree memorizes training noise at high depth without any corresponding gain (in fact, a loss) in generalization.

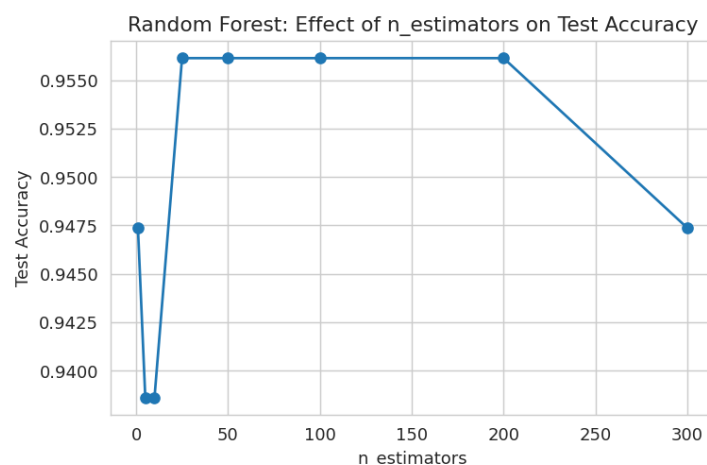


Figure 4: Random Forest: effect of n_estimators on test accuracy

Inference: Test accuracy rises from 0.947 at a single tree to a stable plateau of 0.956 from around 25 trees onward, with no overfitting as n_estimators grows further – unlike Decision Tree depth, adding more trees to a Random Forest does not increase variance, since each additional tree only further averages out (rather than compounds) individual tree error.

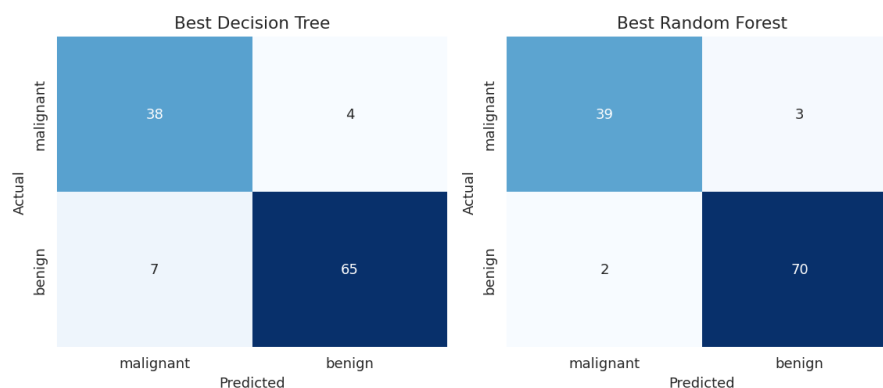


Figure 5: Confusion matrices – best Decision Tree vs. best Random Forest

Inference: Random Forest makes visibly fewer errors of both types (false positives and false negatives) than the Decision Tree on the same test set, most notably fewer missed malignant cases – the clinically more costly error type.

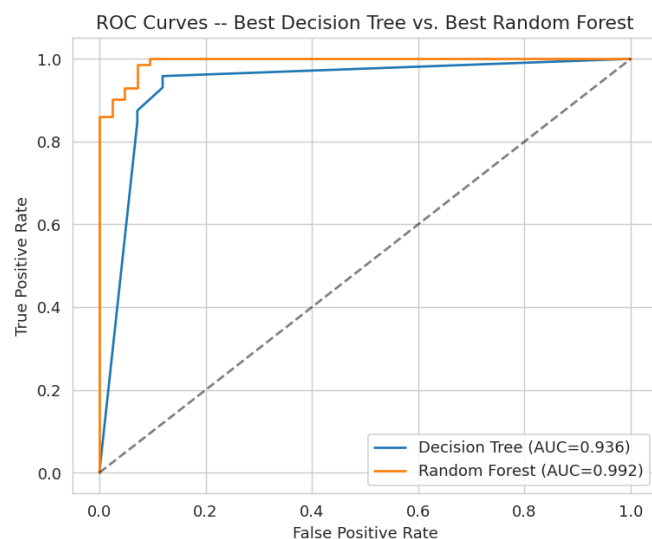


Figure 6: ROC curves – best Decision Tree vs. best Random Forest

Inference: Random Forest’s ROC curve dominates the Decision Tree’s across almost the entire range (AUC 0.992 vs. 0.936), confirming its probability-ranking quality is substantially better, not just its hard classification accuracy.

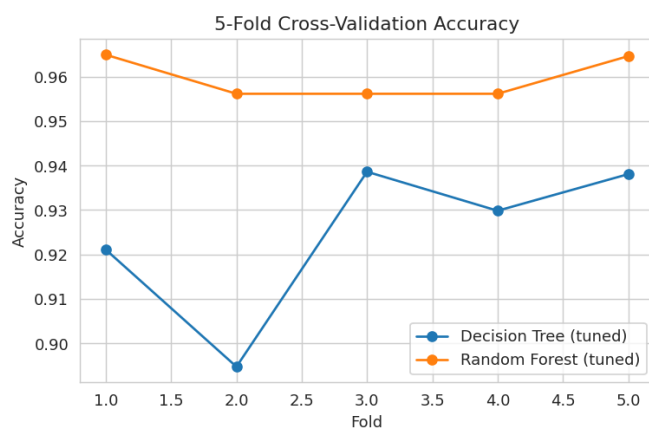


Figure 7: 5-fold cross-validation accuracy – tuned Decision Tree vs. tuned Random Forest

Inference: Random Forest outperforms Decision Tree on every one of the 5 folds (average 0.9596 vs. 0.9245), with visibly less fold-to-fold variance – direct empirical confirmation of bagging’s variance-reduction effect, not just a single-split artifact.

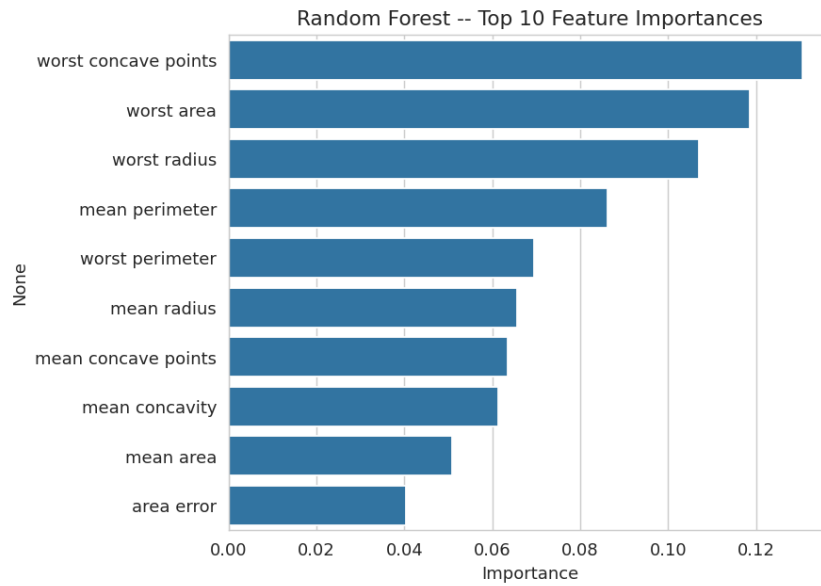


Figure 8: Random Forest – top 10 feature importances

Inference: “Worst concave points”, “worst area”, and “worst radius” are the three most important features by a clear margin, matching the strongest correlations identified in the EDA correlation heatmap (Section 4) and the root/near-root splits of the visualized decision tree below – three independent views of the data agreeing on the same handful of dominant features.

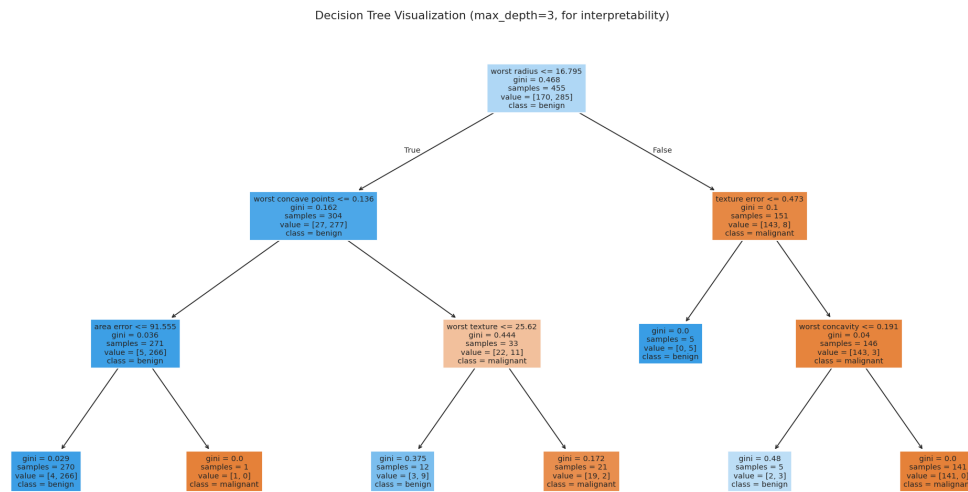


Figure 9: Decision tree visualization (max_depth=3, for interpretability)

Inference: The tree’s root split is on “worst radius” (≤ 16.795), immediately separating a large, highly pure benign-leaning branch (266 of 271 samples benign) from a malignant-leaning branch – illustrating in a single readable diagram exactly why tree-based models are valued for interpretability, a property the Random Forest ensemble trades away for accuracy.

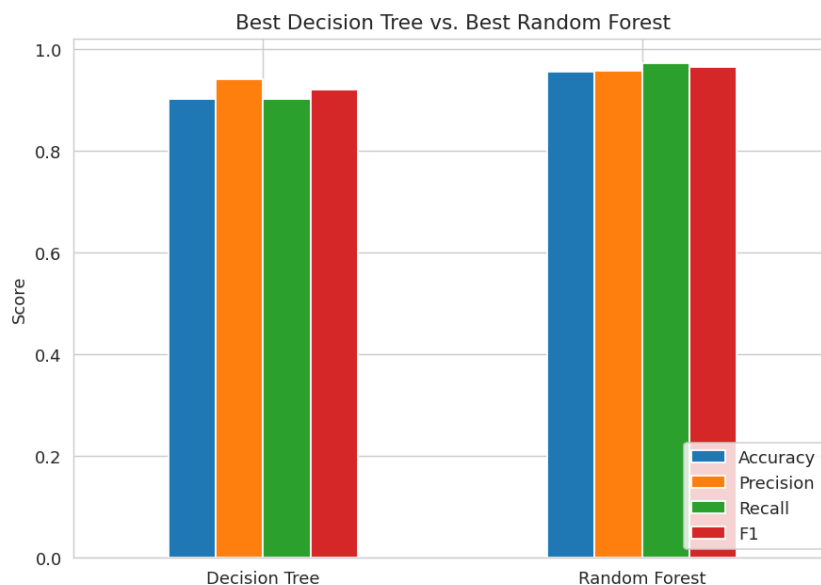


Figure 10: Best Decision Tree vs. best Random Forest – Accuracy / Precision / Recall / F1

Inference: Random Forest leads on every metric, with its largest margin over Decision Tree on Recall (0.972 vs. 0.903) – for a cancer-diagnosis task, this recall gap (fewer missed malignant cases) is arguably the most clinically important difference between the two models.

11 5-Fold Cross-Validation Performance Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	0.9211	0.8947	0.9386	0.9298	0.9381	0.9245
Random Forest	0.9649	0.9561	0.9561	0.9561	0.9646	0.9596

12 Observation Questions

1. **How does tree depth affect overfitting in Decision Trees?** As shown in Figure 7, training accuracy reaches 100% by depth 7 while test accuracy peaks at depth 3–4 and then declines – deeper trees fit training-set-specific noise (individual sample peculiarities) rather than generalizable patterns, the defining signature of overfitting.
2. **Which hyperparameter had the greatest impact on performance?** For the Decision Tree, `max_depth` had by far the largest effect (CV accuracy swinging from 0.921 to 0.939 across depths, with a clear optimum), more than criterion choice or leaf/split minimums. For Random Forest, `max_features` (log2 vs. sqrt) and using an unconstrained `max_depth` together produced the best result, though the overall spread across the Random Forest grid (0.950–0.965) was much narrower than

the Decision Tree's, indicating Random Forest is considerably less hyperparameter-sensitive.

3. **How does Random Forest improve generalization?** By training many trees on bootstrap-resampled data and restricting each split to a random feature subset, individual trees become decorrelated from one another; averaging their predictions cancels out much of each tree's individual overfitting/variance, which is exactly why Random Forest's train-test gap does not widen with more trees (Figure 8) the way the Decision Tree's does with depth (Figure 7).
4. **Did ensemble learning always improve performance? Why or why not?** Yes, in this experiment Random Forest outperformed the tuned Decision Tree on every metric and every cross-validation fold, with no exceptions. This is not a universal guarantee for all datasets, however – ensembling primarily helps when the base learner's error is dominated by variance (as a Decision Tree's typically is); on a dataset where a single tree already generalizes near-optimally with low variance, the ensemble's benefit would be much smaller.

13 Discussion

The Wisconsin Diagnostic Breast Cancer dataset demonstrates textbook Decision Tree overfitting behaviour: unconstrained trees reach perfect training accuracy while test accuracy actually degrades past a shallow optimal depth. Random Forest's bagging and random feature selection directly address this, delivering both higher accuracy (0.956 vs. 0.904) and lower cross-validation variance (Section 14) at a modest training-time cost (0.149s vs. 0.006s – still negligible in absolute terms at this dataset size). The strong agreement between the correlation heatmap, Random Forest feature importances, and the visualized decision tree's early splits (all highlighting “worst radius,” “worst concave points,” and related “worst”-prefixed measurements) gives confidence that both models are learning genuine, clinically-sensible signal rather than spurious correlations. A limitation of this experiment is that hyperparameter search used accuracy as the sole optimization target; for a diagnostic task, optimizing directly for recall (minimizing missed malignant cases) may be more clinically appropriate and could shift the selected hyperparameters.

14 Conclusion

For the Wisconsin Diagnostic Breast Cancer dataset, Random Forest (100 trees, max_depth unconstrained, max_features=log2) is unambiguously the better-performing model, exceeding the tuned Decision Tree on every evaluation metric and every cross-validation fold, most notably on recall – the most clinically consequential metric for a cancer-diagnosis task. The Decision Tree's overfitting at high depth was clearly demonstrated and corrected via hyperparameter tuning (optimal depth ≈ 5), but even the best-tuned single tree could not match the ensemble's generalization. The trade-off is a loss of the single tree's direct visual interpretability, partially offset by Random Forest's feature-importance ranking, which converges on the same dominant features the tree itself split on first.

15 References

- [1] Scikit-learn Documentation, “Decision Trees,” <https://scikit-learn.org/stable/modules/tree.html>
- [2] Scikit-learn Documentation, “Random Forests,” <https://scikit-learn.org/stable/modules/ensemble.html#forest>
- [3] UCI Machine Learning Repository, “Breast Cancer Wisconsin (Diagnostic) Data Set,” <https://archive.ics.uci.edu>